Numerical Simulation of Quantum Size Effects

Ning Siyuan

10/11/2022

Methods

The Finite Difference Method

$$\frac{df}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x-h)}{2h}$$

$$\frac{df}{dx} = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 & 0 \\ -1 & 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & -1 & 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & -1 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & 0 & \dots & -1 & 0 \end{bmatrix} \begin{bmatrix} f(a) \\ f(a+h) \\ f(a+2h) \\ f(a+3h) \\ \dots \\ f(a+(n-2)h) \\ f(b) \end{bmatrix} /(2h) = D|f\rangle$$

$$\frac{d^2 f}{dx^2} = \lim_{h\to 0} \frac{f(x-h) - 2f(x) + f(x+h)}{h^2}$$

$$\frac{d^2 f}{dx^2} = \begin{bmatrix} -2 & 1 & 0 & 0 & \dots & 0 & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 & 0 \\ 0 & 0 & 1 & -2 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & -2 & 1 \\ 0 & 0 & 0 & 0 & \dots & 1 & -2 \end{bmatrix} \begin{bmatrix} f(a) \\ f(a+h) \\ f(a+2h) \\ f(a+3h) \\ \dots \\ f(a+(n-2)h) \\ f(b) \end{bmatrix} /h^2 = \mathrm{Lap}|f\rangle$$

Methods

Differential Equation → Matrix Equation

Express the Hamiltonian as a matrix operator

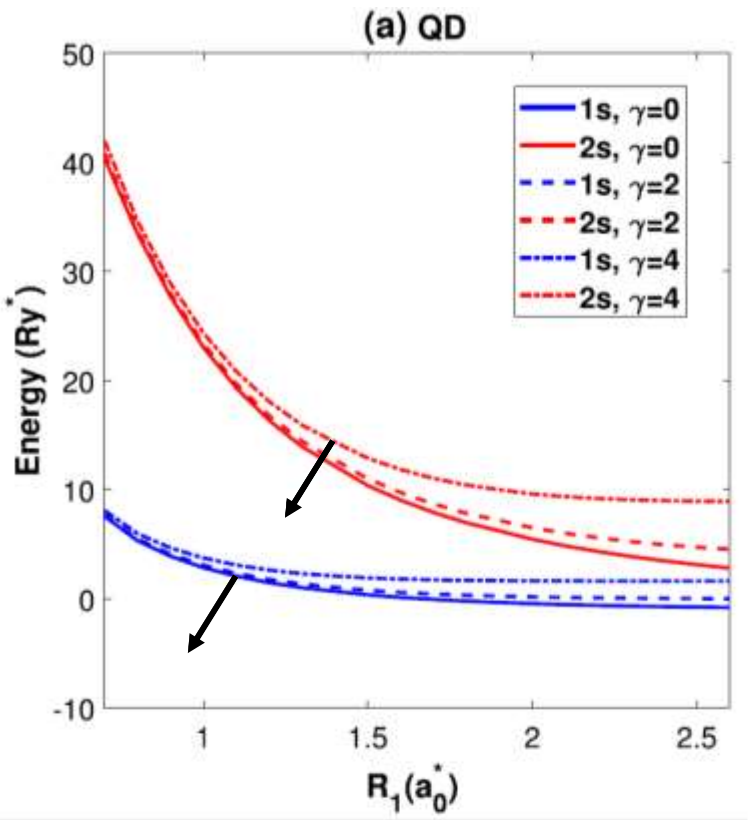
$$\hat{H}|\psi\rangle = E|\psi\rangle \qquad [\hat{K} + \hat{V}]|\psi\rangle = E|\psi\rangle \qquad \left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right]|\psi\rangle = E|\psi\rangle$$

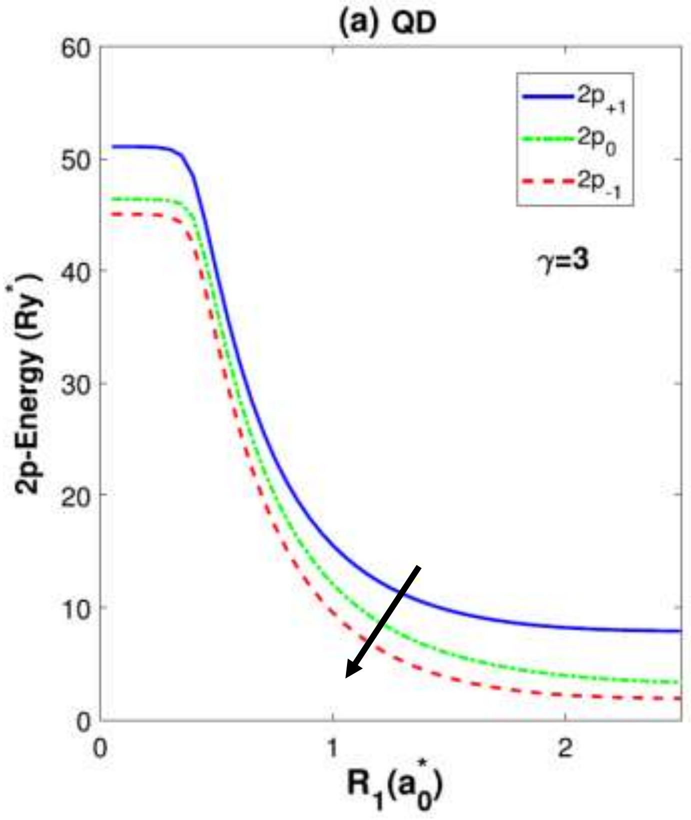
$$|\psi\rangle = \begin{bmatrix} \psi(x_1) \\ \psi(x_2) \\ \vdots \\ \psi(x_n) \end{bmatrix} \qquad \hat{K} = -\frac{\hbar^2}{2m\ell^2}\begin{bmatrix} -2 & 1 & 0 & \dots & \dots & 1 \\ 1 & -2 & 1 & \dots & \dots & 0 \\ 0 & 1 & -2 & \dots & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & 1 \\ 1 & 0 & 0 & \dots & 1 & -2 \end{bmatrix} \qquad V(\hat{x}) = \begin{bmatrix} V(x_1) & 0 & \dots & \dots & \dots & 0 \\ 0 & V(x_2) & 0 & \dots & \dots & 0 \\ \vdots & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & 0 \\ 0 & 0 & \dots & \dots & 0 & V(x_n) \end{bmatrix}$$

Results

① $GaAs/Ga_{1-x}Al_xAs$ core/shell quantum dots

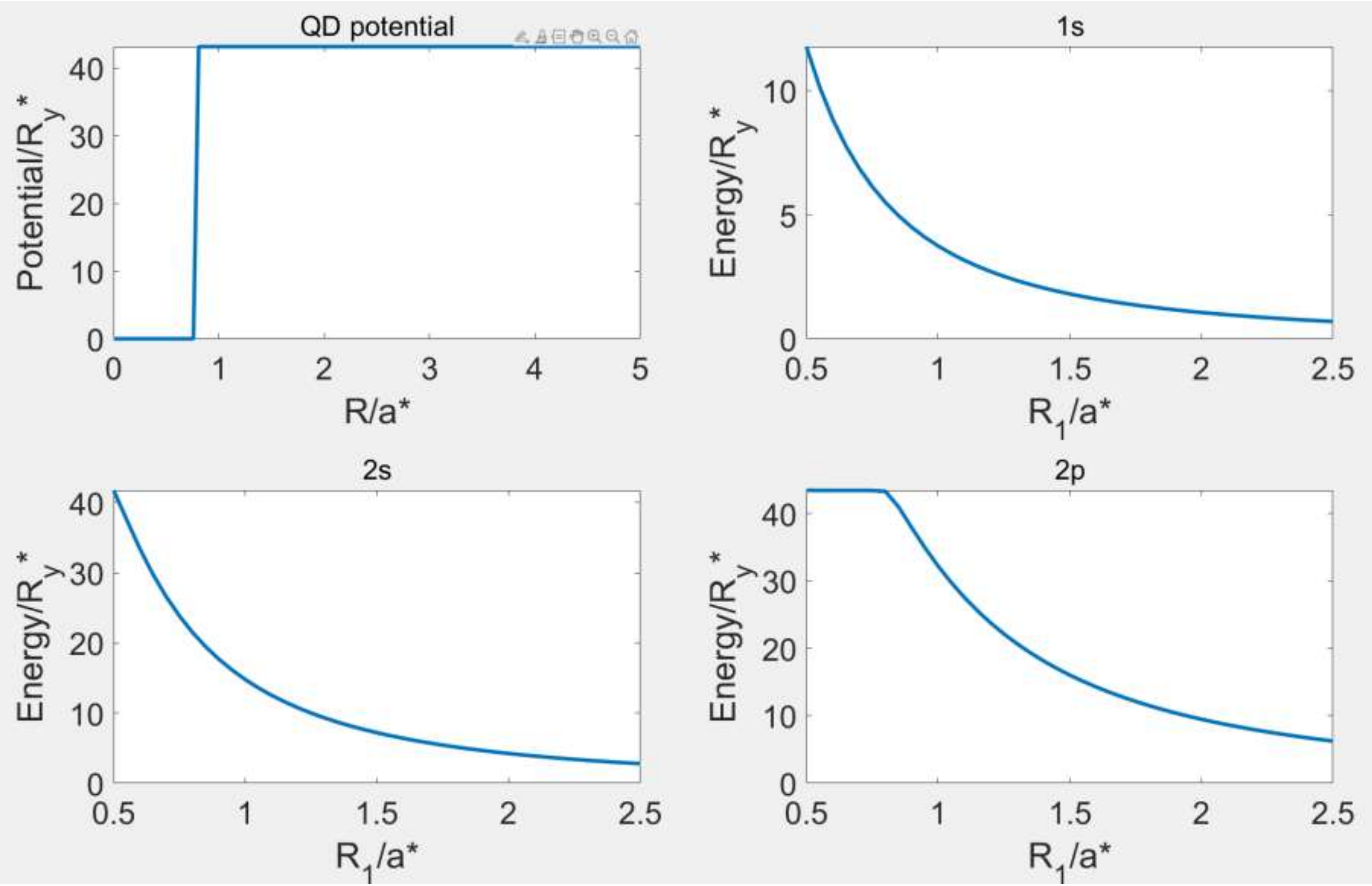
$$V_c^{QD}(r) = \begin{cases} 0 & r \le R_1 \\ V_0 & R_1 < r < R_2 \\ \infty & otherwise \end{cases}$$





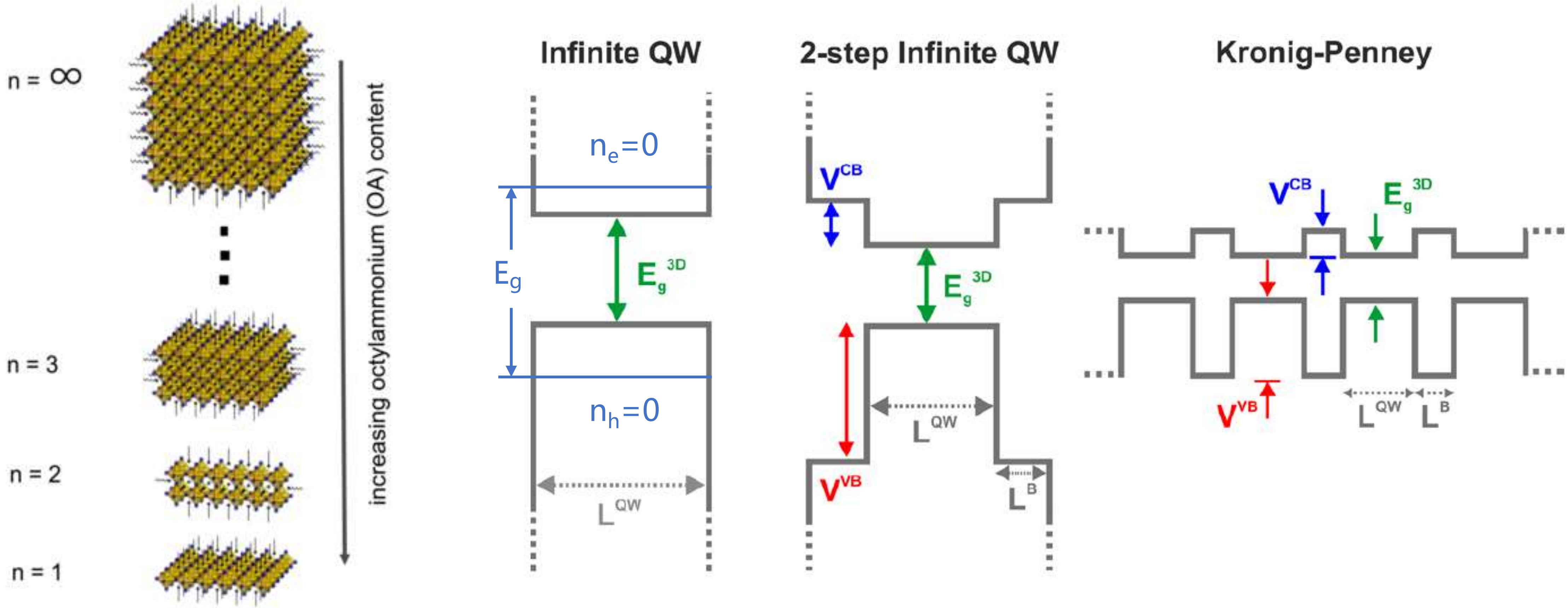
Rahimi, F. *et al. Opt Quant Electron* **53**, 47 (2021).

Results


QD potential
Potential/R_y^*
R/a*
0
10
20
30
40
0
1
2
3
4
5
1s
Energy/R_y^*
R_1/a*
0
5
10
0.5
1
1.5
2
2.5
2s
Energy/R_y^*
R_1/a*
0
10
20
30
40
0.5
1
1.5
2
2.5
2p
Energy/R_y^*
R_1/a*
0
10
20
30
40
0.5
1
1.5
2
2.5

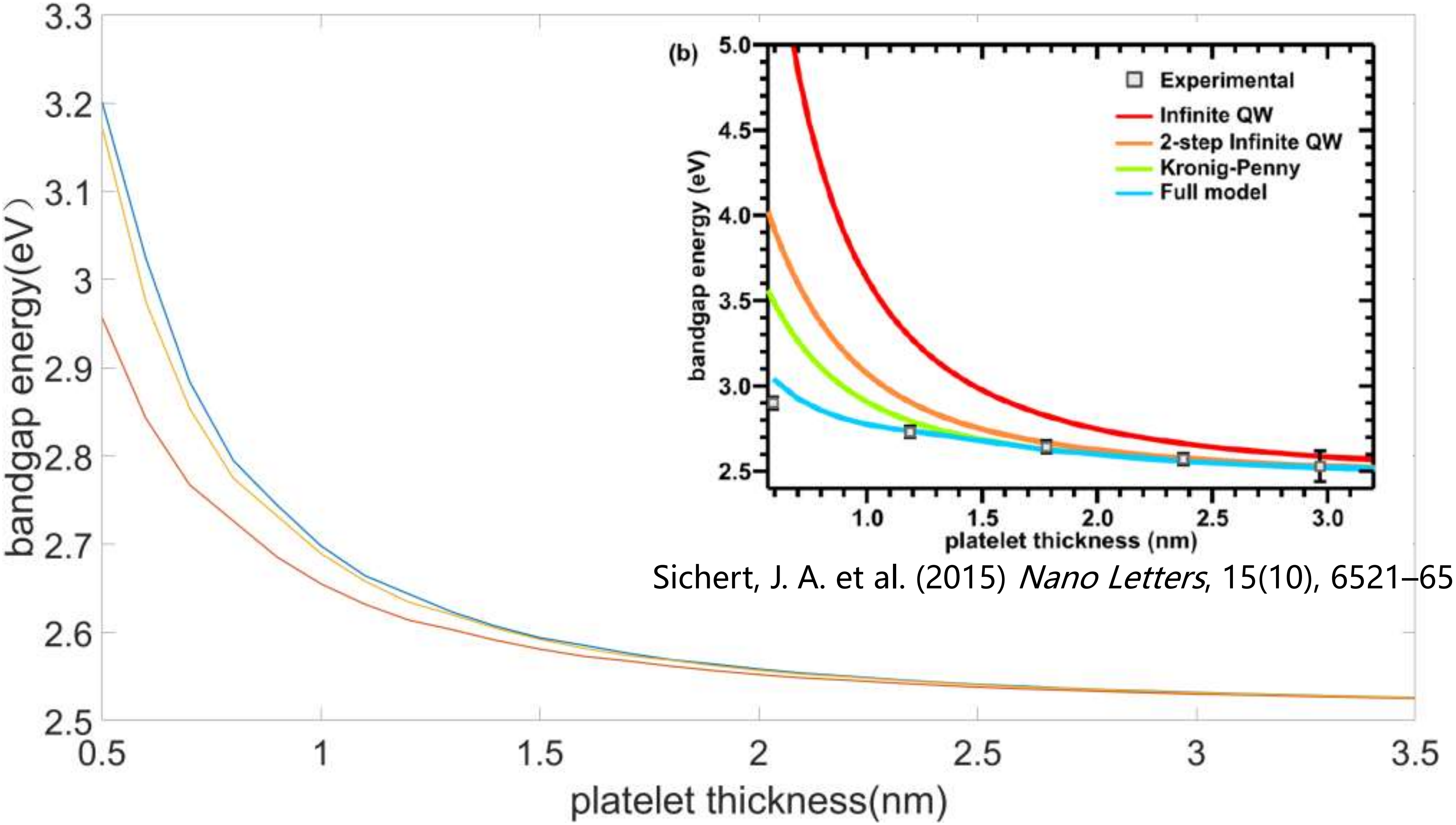
Results

② Methylammonium lead bromide ($MAPbBr_3$) Nanoplatelets



$$E_g = E_g^{3D} + E_e + E_h$$

Results



Sichert, J. A. et al. (2015) *Nano Letters*, 15(10), 6521–6527.